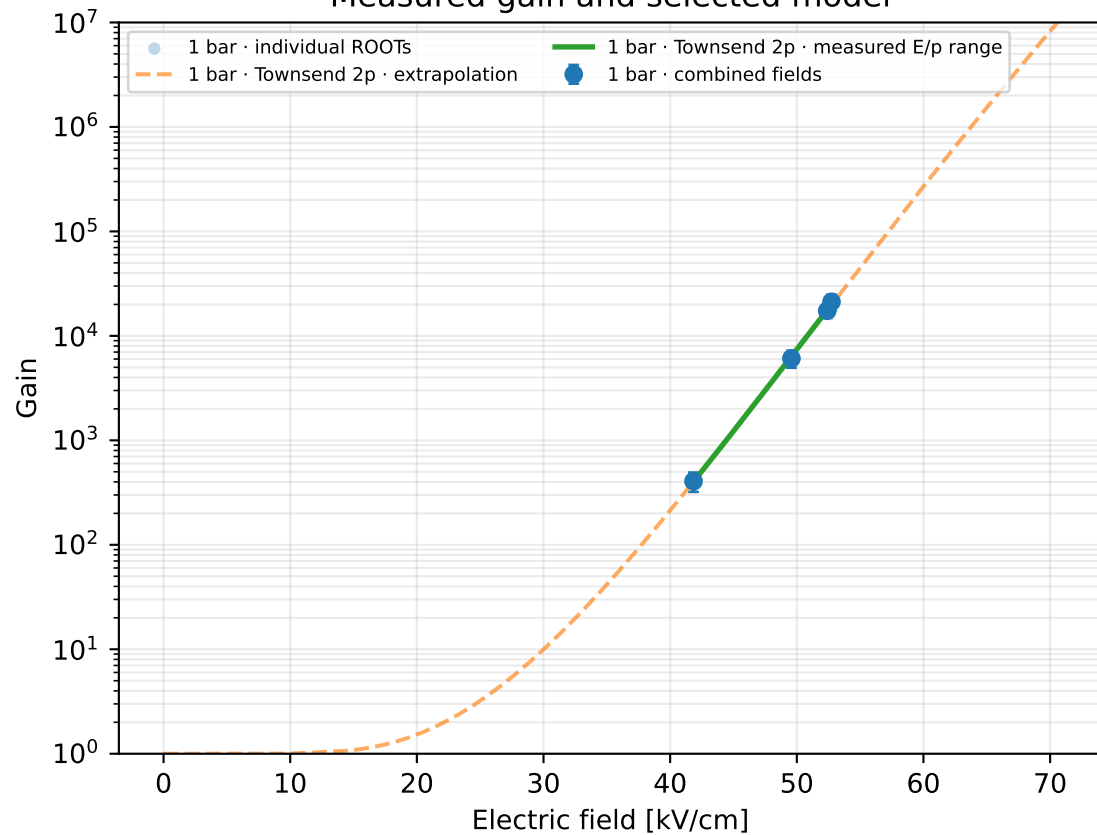
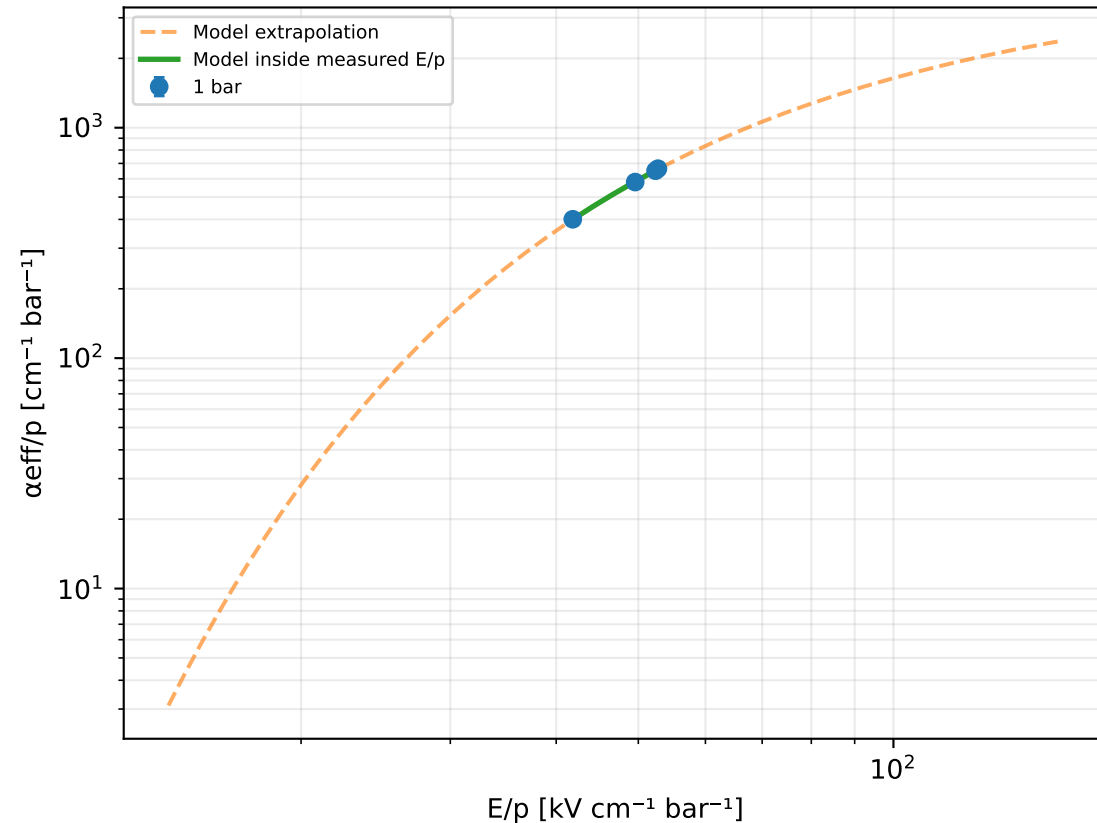


ArIsoCF4 · Ar 78 % / Iso 20 % / CF4 2 % · gap 0.15 mm

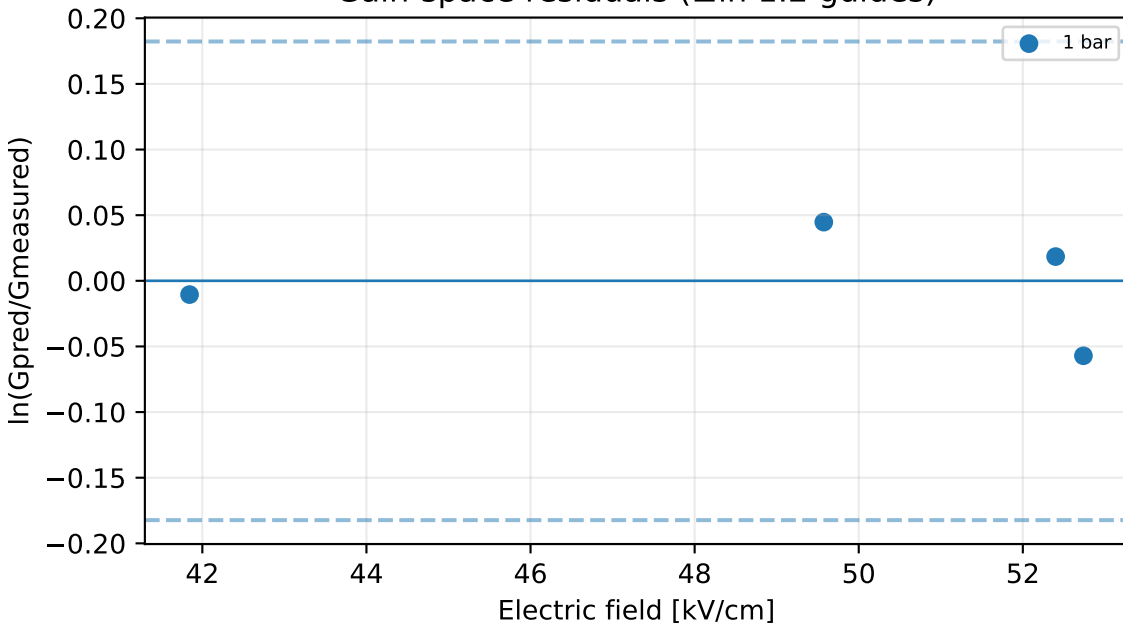
Measured gain and selected model



Reduced Townsend fit

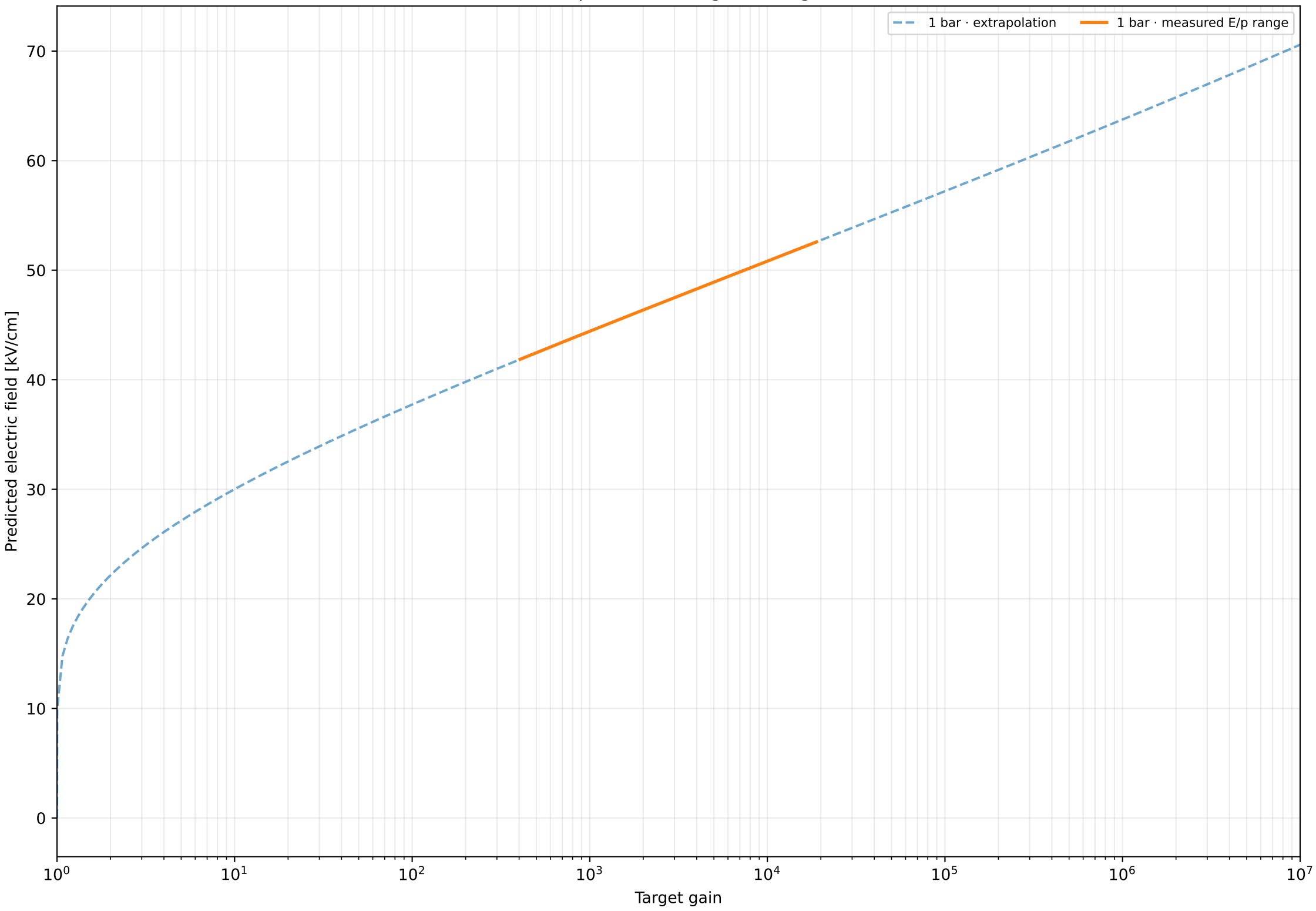


Gain-space residuals ($\pm \ln 1.2$ guides)



ROOT runs: 4
 Unique physical points: 4
 Pressures: 1 bar
 Selected model: Townsend 2p
 Status: rejected_poor_cross_validation
 Training RMSE: $\times 1.004$
 Cross-validation median: unavailable
 Cross-validation worst: unavailable
 Reduced-field range: 41.84–52.74 kV cm⁻¹ bar⁻¹
 A=4534.3, B=101.62, m=0, n=1
 Covariance condition: 777
 Warnings: poor_cross_validation, large_cross_validation_outlier

Field extrapolation from gain 1 to gain 10^7



p [bar]	E [kV/cm]	E/p	Gain	σ Gain	α_{eff} [cm ⁻¹]	npe	runs
1	41.8443	41.8443	406.45	85.4	400.497	20	1
1	49.5743	49.5743	6080.9	1.12e+03	580.861	20	1
1	52.3998	52.3998	17372.4	2.53e+03	650.843	20	1
1	52.7386	52.7386	21180.05	3.64e+03	664.054	20	1

Progressive model selection

The higher-order model is selected only when new points improve out-of-sample prediction.

Candidate	k	AICc	CV median	CV worst	usable	warnings
Townsend 2p	2	-28.1	—	—	no	poor cross validation, large cross validation outlier